

GRAPH-REGULARIZED FEDERATED LEARNING FOR STATION-LEVEL WEATHER PREDICTION USING FMI OPEN DATA

ABSTRACT

We study station-level weather prediction using open data from the Finnish Meteorological Institute (FMI) within a federated learning (FL) framework. Each FMI weather station is modeled as one node in an undirected weighted graph, where edges represent inter-station similarity derived from geographic proximity or from a combination of geographic proximity and training-period temperature correlation. Local supervised datapoints are constructed from daily weather measurements and seasonal features, and the learning task is next-day temperature prediction. We formulate the problem as a graph-regularized variation minimization problem with local squared-error losses and implement a synchronous fixed-point style FL algorithm for solving the resulting optimization problem. Two graph-based FL systems are evaluated and compared against local-only and pooled global linear baselines on a multi-station FMI dataset with chronological train-test splitting. The study examines whether graph-based personalization improves predictive performance in this setting and analyzes the effect of graph construction on the resulting FL system.

Index Terms— Federated learning, networks, personalized machine learning, trustworthy artificial intelligence

1. INTRODUCTION

Federated learning (FL) is well suited to learning problems in which data are naturally distributed across multiple entities and should not be treated as one homogeneous dataset. In this project, the raw data source is the Finnish Meteorological Institute (FMI) Open Data service, and each node in the FL network corresponds to one FMI weather station. This makes station-level weather prediction a natural FL application: each station collects its own local observations, while nearby stations are related through shared meteorological structure. At the same time, stations are not identical, since local climate, geography, and seasonal effects can differ across locations. This motivates graph-based FL methods that allow stations to share information without enforcing a fully shared global model.

We study next-day temperature prediction from daily station observations. For each station, local datapoints are constructed from daily weather variables and seasonal features,

and the goal is to predict the next-day mean temperature. We consider graph-regularized personalized FL, where each station learns its own local model and neighboring stations are coupled through an undirected weighted graph. This follows the project workflow of modelling the application as a graph-based variation minimization problem and deriving an iterative FL algorithm from the corresponding fixed-point characterization.

The application domain is weather forecasting from geographically distributed sensors. A pooled global model can exploit common large-scale structure across stations, whereas local models can adapt to site-specific variation. Graph-based FL provides an intermediate alternative by coupling related stations according to an empirical graph, for example based on geographic proximity or similarity in observed temperature dynamics. In this work, we compare two such FL systems: one using geography-only edges and one combining geography with training-period temperature correlation.

The remainder of this report is organized as follows. Section 2 defines the FL problem, including the station graph, local datapoints, local model, and loss function. Section 3 presents the graph-regularized learning methodology and the corresponding synchronous fixed-point style update. Section 4 reports numerical experiments comparing local, global, and graph-based FL systems on FMI station data. Section 5 concludes the report and discusses limitations and possible future improvements.

2. PROBLEM FORMULATION

We use FMI Open Data and model each FMI weather station as one node in the FL network. After filtering for sufficient data coverage and required variables, the final dataset contains 42 stations with daily observations from 2023-01-01 to 2024-12-31. Stations were selected to ensure that each node has enough training and test samples for meaningful local learning and evaluation.

We represent the FL system by an undirected weighted graph $G = (V, E, A)$. We study two graph constructions. The first uses geographic k -nearest neighbours, with edge weights defined by a Gaussian kernel of inter-station distance. The second uses the same geographic neighbourhoods but multiplies the geographic weight by the positive training-period correlation between station temperature series. Thus, the first

graph is geography-driven, while the second combines geographic and data-driven similarity.

For station i and day t , the local feature vector is

$$x_{i,t} = [tday, tmin, tmax, snow, \sin(doy), \cos(doy)],$$

and the label is the next-day mean temperature,

$$y_{i,t} = tday_{i,t+1}.$$

Each station trains a local linear model with parameter vector $w_i \in \mathbb{R}^d$. The local loss function is the empirical squared-error loss

$$L_i(w_i) = \frac{1}{m_i} \|X_i w_i - y_i\|_2^2.$$

We use linear models because they are interpretable, computationally simple, and well matched to graph-regularized optimization.

3. METHODOLOGY

We use a smooth graph-regularized FL objective

$$\min_{w_1, \dots, w_n} \sum_{i=1}^n L_i(w_i) + \alpha \sum_{\{i,j\} \in E} A_{ij} \|w_i - w_j\|_2^2.$$

Here, $\|w_i - w_j\|_2^2$ is the chosen variation measure, and α controls the strength of graph coupling.

The local loss at station i is the squared-error regression loss

$$L_i(w_i) = \frac{1}{m_i} \|X_i w_i - y_i\|_2^2.$$

The stationwise optimality condition gives the synchronous iterative update

$$w_i^{(t+1)} = w_i^{(t)} - \eta \left[\nabla L_i(w_i^{(t)}) + 2\alpha \sum_{j \in \mathcal{N}(i)} A_{ij} (w_i^{(t)} - w_j^{(t)}) \right].$$

Thus, each node updates its local model using its own data and the current model parameters received from neighboring stations. This yields a message-passing FL algorithm in which raw measurements are not shared.

We compare two FL systems with the same local model and optimization method but different graph constructions: a geography-only graph and a geography+correlation graph. We also implemented a non-smooth TV-style variant, but the main experiments focus on the smooth squared- ℓ_2 formulation because it was more stable in our setting.

4. NUMERICAL EXPERIMENTS

We evaluate the proposed FL methods on daily FMI weather observations from 42 stations. The data cover the period from

2023-01-01 to 2024-12-31. For each station, the target is next-day mean temperature, and the feature vector is

$$x_{i,t} = [tday, tmin, tmax, snow, \sin(doy), \cos(doy)],$$

where doy denotes the day-of-year. Thus, each local data-point corresponds to one station-day pair, and the label is

$$y_{i,t} = tday_{i,t+1}.$$

Since the data are time series, we use a chronological split. Data from 2023 are used for training, and data from 2024 are used for testing. This avoids information leakage from future observations into past model fitting. Features are standardized using training data only, and the same transformation is then applied to the test data. A constant intercept term is appended after standardization.

We compare four systems:

- **Local baseline:** each station trains an independent linear regression model using only its own local data.
- **Global baseline:** one pooled linear regression model is trained on the union of all station training data.
- **FL System A (graph-geo):** a graph-regularized FL model using a geography-only graph.
- **FL System B (graph-geo-corr):** a graph-regularized FL model using a graph that combines geographic proximity with training-period temperature correlation.

The two FL systems differ in a structural design choice, namely the graph construction, which satisfies the requirement to compare at least two distinct FL systems.

For both FL systems, the local model at each node is linear, and the optimization method is the synchronous fixed-point style update described in Section 3. The regularization parameter α is selected using a sweep over candidate values

$$\alpha \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5, 1.0\}.$$

The learning rate is set to $\eta = 10^{-3}$, the number of training iterations is 3000 for the parameter sweep and 5000 for the final selected model, and the local parameters are initialized to zero. For the graph-regularized runs, the geographic graph uses k -nearest neighbours with $k = 3$ and Gaussian distance weights with bandwidth $\sigma = 300$ km. In the correlation-weighted graph, the geographic weight is multiplied by the positive training-period Pearson correlation between station temperature series.

The main evaluation metric is root mean squared error (RMSE) on the 2024 test set, computed separately for each station and then averaged across stations. We also report whether each graph-based system outperforms the local and global baselines on a station-by-station basis.

Table 1 summarizes the main results. The pooled global model achieves the best mean RMSE, while both graph-based

Table 1: Mean test RMSE across 42 stations.

Model	Mean RMSE
Local	3.109
Global	2.905
Graph-geo	3.119
Graph-geo-corr	3.119

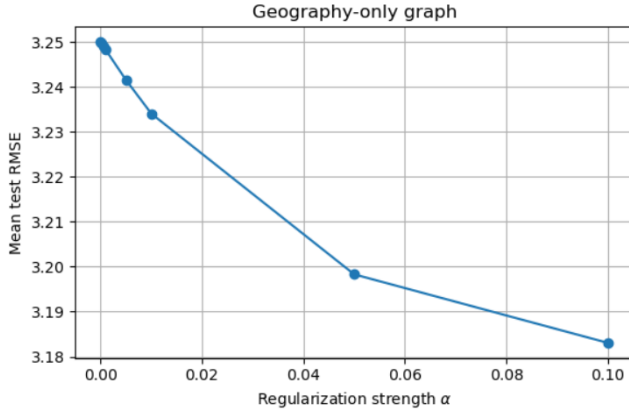


Fig. 1: Mean test RMSE versus regularization strength α for the geography-only graph.

FL systems underperform both the global and local baselines. The two graph constructions produce very similar average performance.

For the geography-only graph, the best result in the regularization sweep is obtained at $\alpha = 1.0$, with mean RMSE 3.205 in an intermediate tuning run and 3.119 in the final run with the updated feature set and intercept handling. For the geography+correlation graph, the best value is also $\alpha = 1.0$, yielding mean RMSE 3.170 in the sweep and 3.119 in the final comparison. Although the graph-based systems improve as α increases, neither surpasses the pooled global baseline.

We also evaluate per-station win rates. The geography-only graph outperforms the local baseline on 2.5% of stations and the global baseline on 0% of stations. The geography+correlation graph yields the same fractions. Thus, the graph-based personalized models are not competitive for the present task under the current feature design and graph construction.

Fig. 1 and Fig. 2 show the effect of the regularization parameter α for the two FL systems. In both cases, larger α values improve performance relative to very small values, but the final results remain inferior to the pooled global model. Fig. 4 shows predictions versus actual values for next-day temperature at a representative station, illustrating that the selected model captures the overall trend of the target variable.

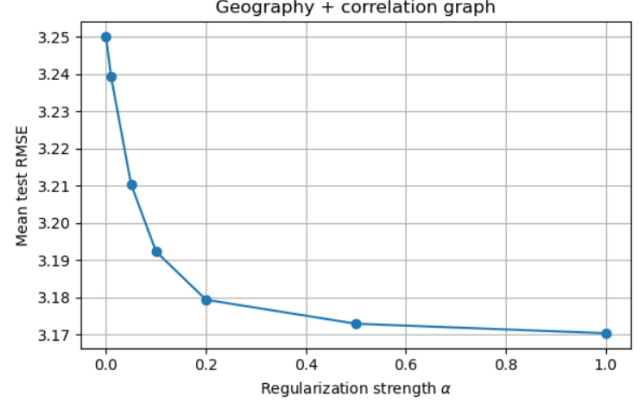


Fig. 2: Mean test RMSE versus regularization strength α for the geography+correlation graph.

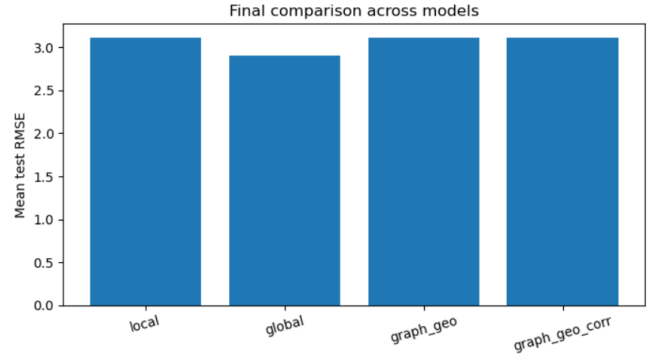


Fig. 3: Final comparison of mean test RMSE across the local, global, geography-only, and geography+correlation systems.

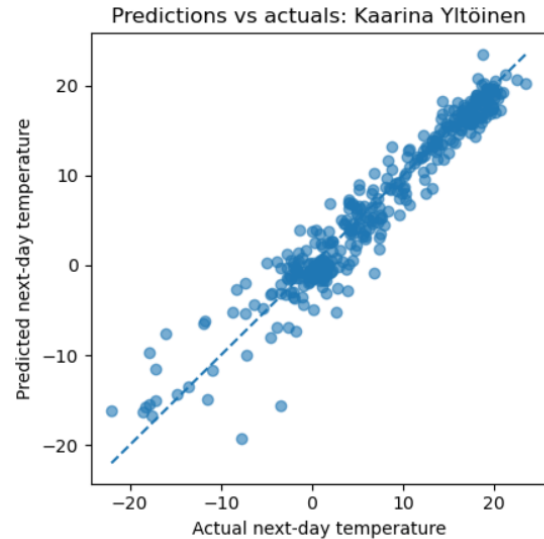


Fig. 4: Predictions versus actual values for next-day temperature at Kaarina Yltöinen under the selected model.

5. CONCLUSION

We formulated station-level weather prediction from FMI Open Data as a graph-based federated learning problem in which each weather station is one node in an empirical graph. We compared two graph-based FL systems, one using geography-only edges and one using geography+correlation edges, against local-only and pooled global linear baselines.

On the 42-station next-day temperature prediction task, the pooled global model achieved the best mean test RMSE, while both graph-based FL systems underperformed both the global and local baselines. Thus, under the present feature design and graph construction, graph-based personalization did not improve predictive performance. This suggests that the task is dominated by shared large-scale structure that is already captured well by a pooled linear model.

The study nevertheless provides a complete FL pipeline on real FMI data, including graph construction, local model definition, graph-regularized optimization, and comparison of multiple FL systems. Limitations include the simple local model, the restricted feature set, and the chosen graph constructions. Future work could explore richer features, alternative similarity graphs, and more expressive local models.

6. REFERENCES

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